Planetary Bean Bouncer – Mechanical Build Guide

Before you begin:
- Review the full guide once before assembling.
- Print **Test_1.stl** to verify hole tolerances for screws, nuts, and bearings.
- Print **Gearing_5_1_16_24_64_Test_Rev36.stl** to test planetary gear meshing (can be used as a fidget after printing).
- Verify you have all hardware listed in the BOM.
- Do not fully tighten fasteners until shaft alignment is confirmed.

This guide documents the mechanical assembly of the Planetary Bean Bouncer coffee roaster. Electronics wiring and configuration are not covered in this document and will be addressed in a separate guide.



Frame

The aluminum extrusion part of the frame consists of 4-2040x500mm and 2-2040x140mm V-Slot extrusion. You should also be able to substitute T-Slot 2040 if you prefer, but I have not tested those. I am using 74 Tee Nuts and all the 3D printed parts are sized to use M5x12mm screws. I use 18-8 stainless steel button head screws as I prefer them. Anything outside the sifter does not need to be stainless though. You might have less or more of these Tee Nuts depending on how you configure it. For instance, you will probably power the 12 V supply for the motor differently than I do or you might not want to bother with the planetary gear system.

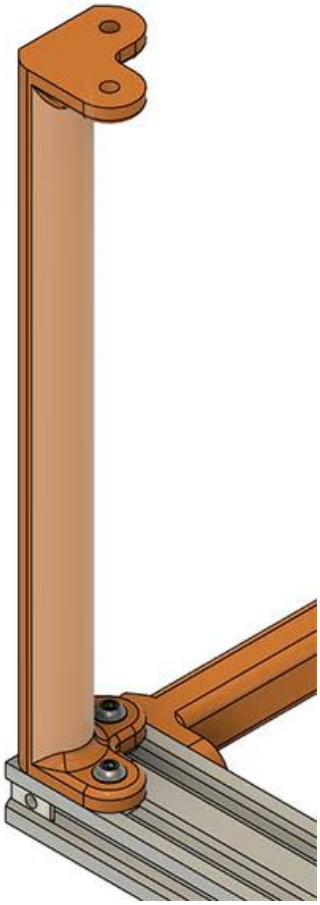
We will start with building the bottom frame. For the bottom frame we start by assembling **V2040_Bot_Connect_1 (**1st part of **2040_Bot_Connects.stl)** and the **Heatgun_Center_V2 3D** prints. If you are using a different heat gun than in BOM, this probably will not work for you. The Heat gun center part is not necessary but makes it easy to line up the heat gun if you are using the same heat gun I have listed in BOM. You need 4-M5x12mm screws with Tee Nuts for the Heat gun center part and 2-M5x12mm screws with Tee Nuts for the **V2040_Bot_Connect_1**



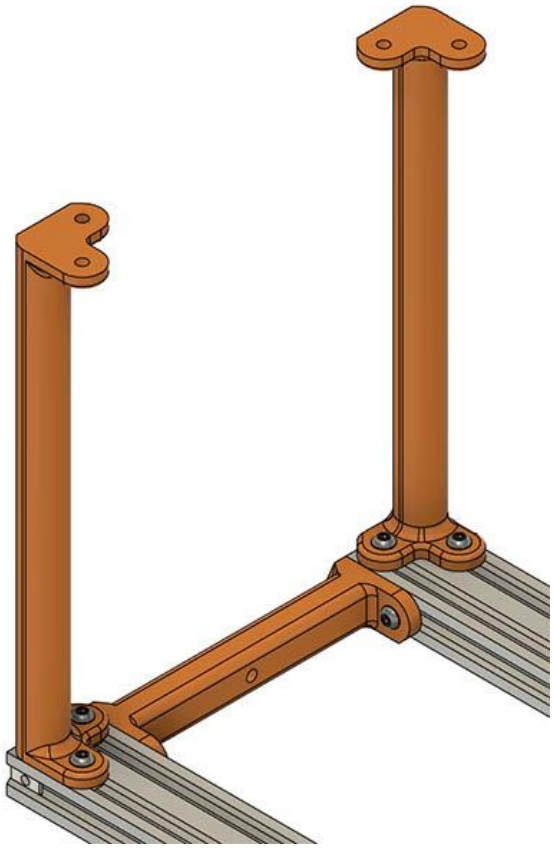
Connect **V2040_Bot_Connect_**2 **(**2nd part of **2040_Bot_Connects.stl)** at each end of the frame with 2-M5x12mm screws and Tee Nuts like the other end.



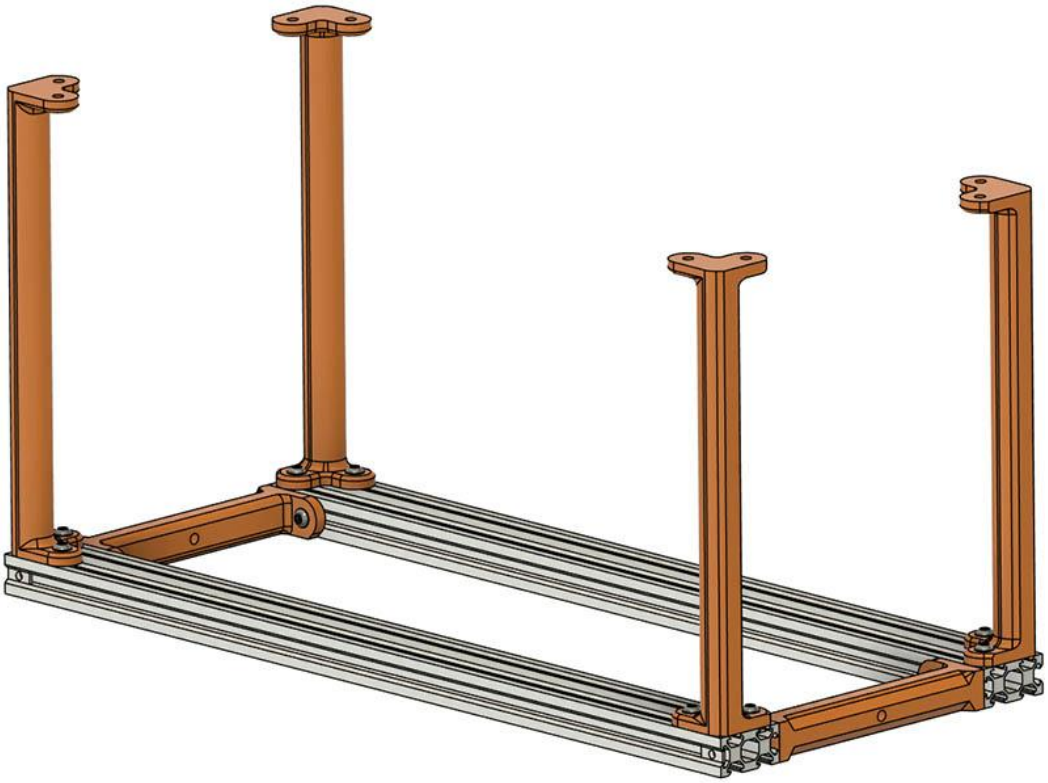
Add **V2020_Column_1 (**1st part of **2020_CornerColumns.stl)** to the front corner below where sifter will be using 2-M5x12mm screws. These columns are slightly different because of print orientation, but essentially any of the 4 can go in any corner.



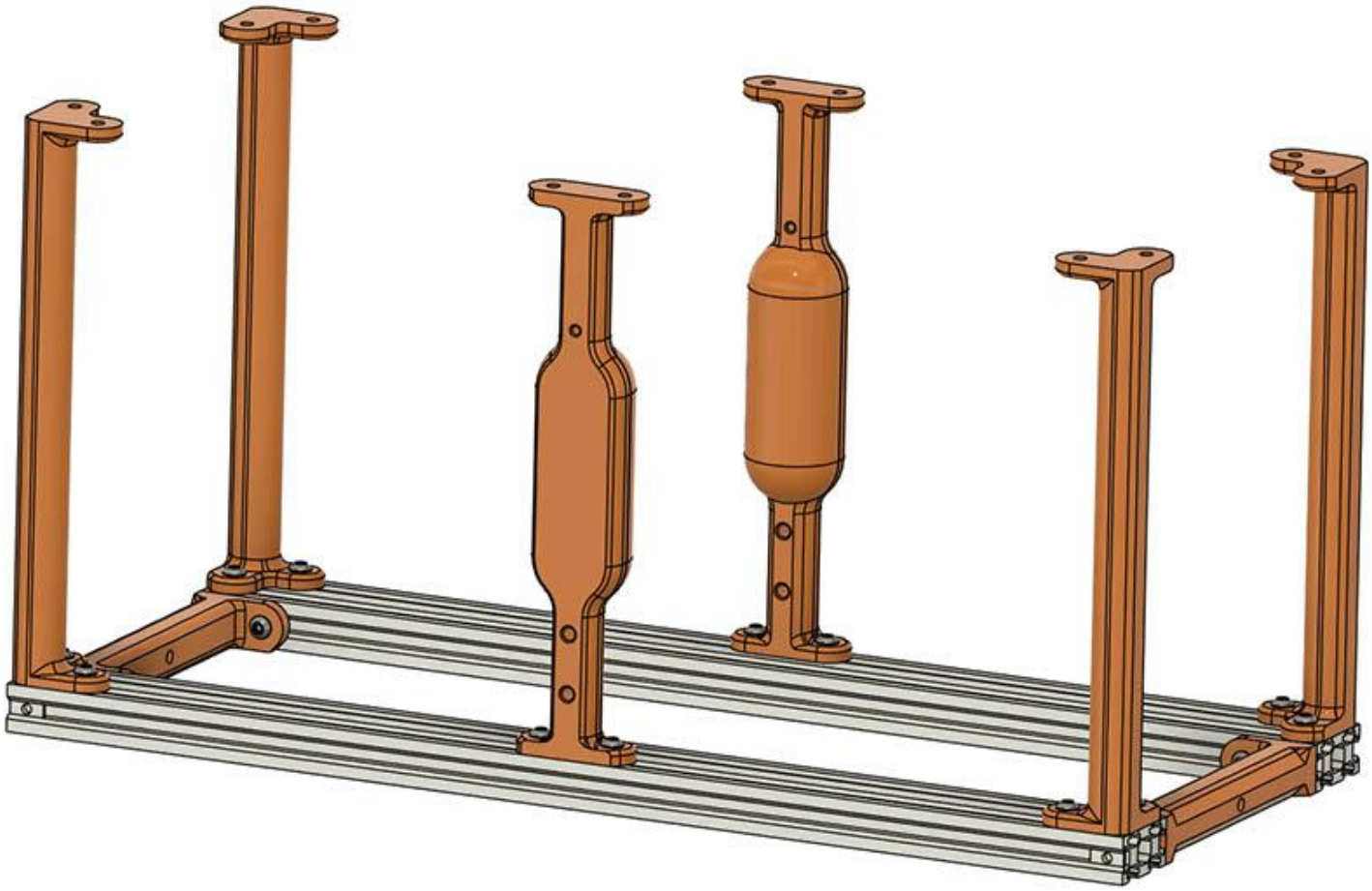
Add **V2020_Column_2 (**2nd part of **2020_CornerColumns.stl)** to opposite short side using 2-M5x12mm screws.



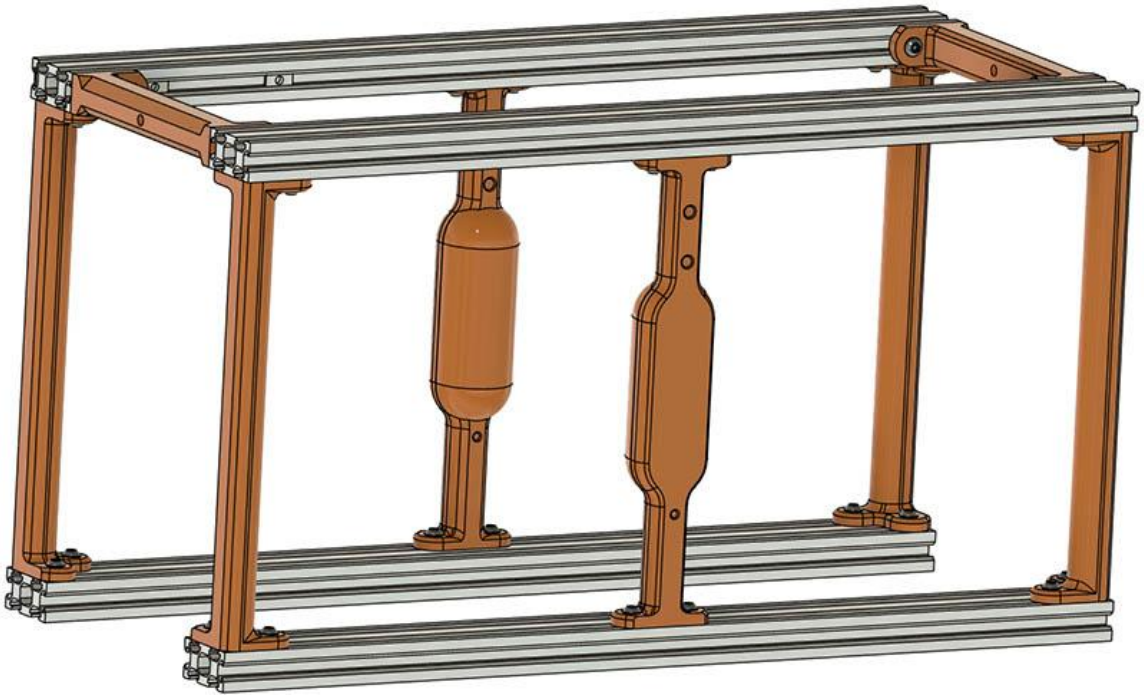
Do the same for the opposite side using 4-M5x12mm screws.



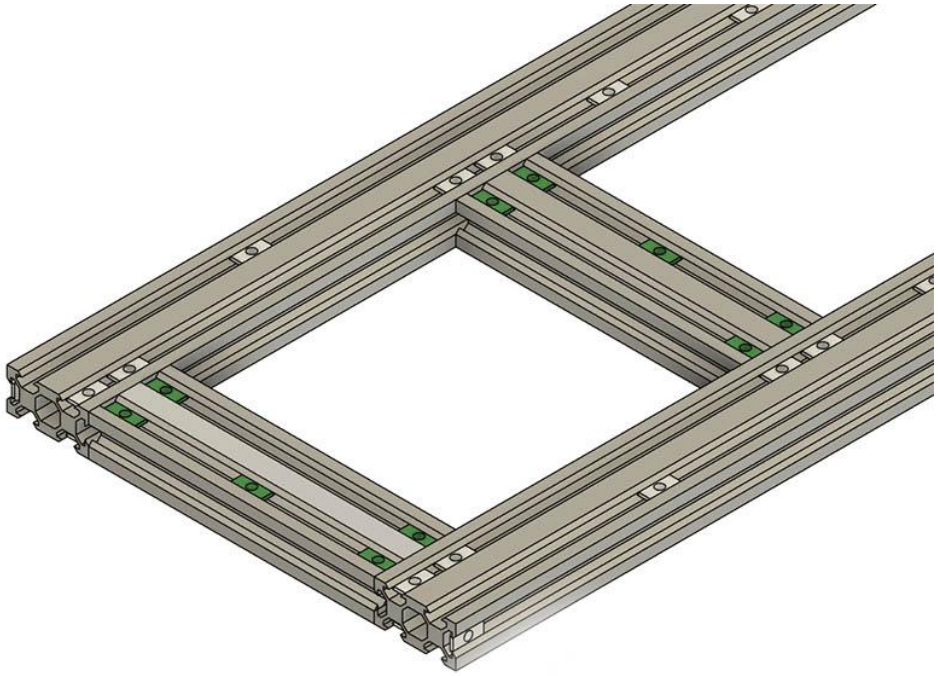
Assemble **CenterColumn_1.stl** and **CenterColumn_2.stl** to the bottom frame using a total of 4-M5x12mm screws as shown. They do not necessarily have to go in the middle. I have the back one in the middle for grabbing the center to dump roasted beans, but the front one I have closer to electronics to fasten MAX6675 and wire tie motor wire.



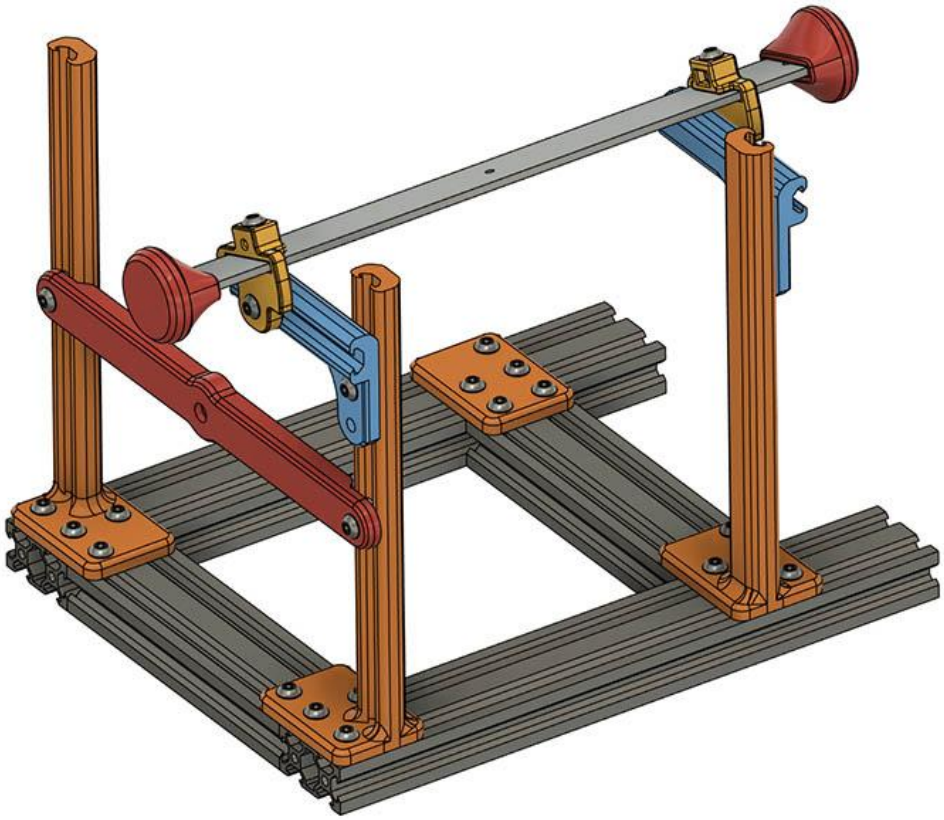
Turn the bottom frame upside down and assemble the 2-2040x500mm lengths to it using 12-M5x12mm screws with a Tee Nut for each screw.



Turn frame right side up again and add 10 Tee Nuts as shown in green to the 2–2040x140mm length extrusions. They do not need to be positioned correctly yet.



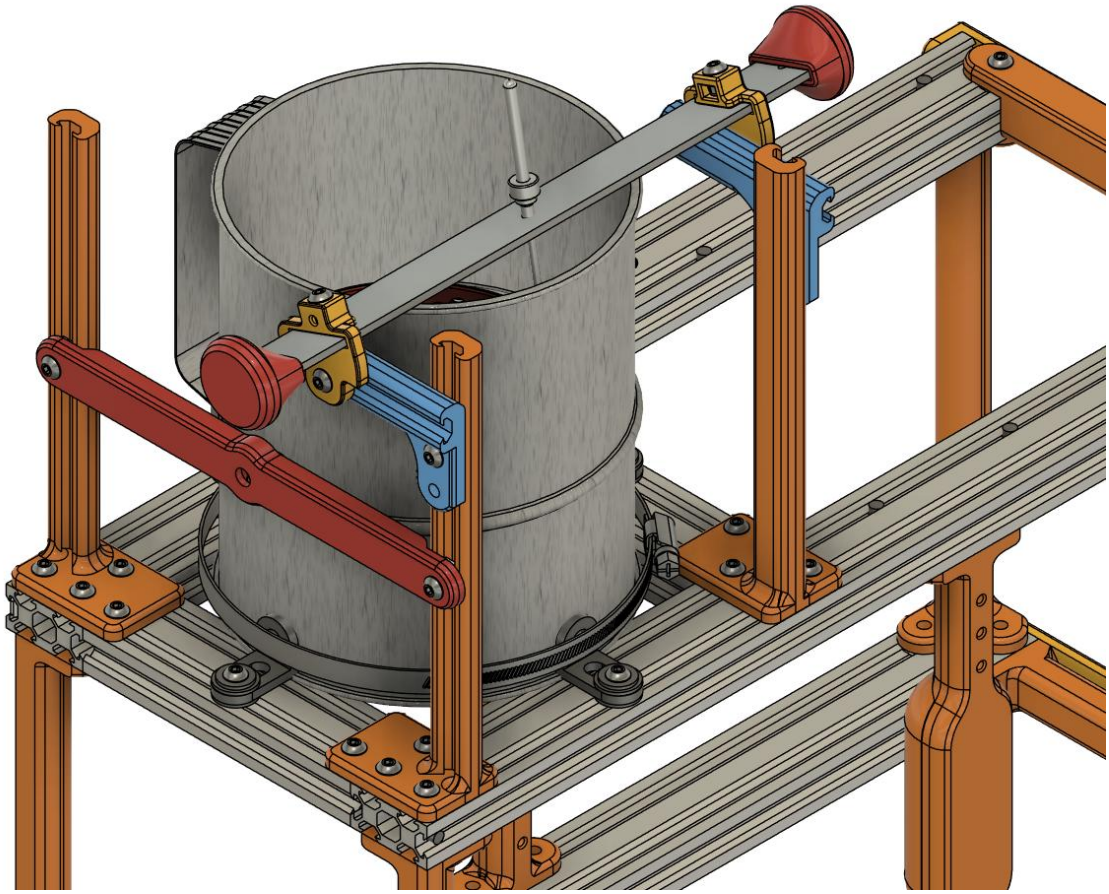
Attach **Bearing_Column_1.stl** and **Bearing_Column_2.stl** (from **BearingColumns.stl**) to the end of the 2040x500mm top frames and the short 2040x140mm extrusion as shown. Make sure you have a Tee Nut as shown in the middle of short 2040. It does not have to be centered yet. Use 8-M5x12mm screws with Tee Nuts in the extrusions to connect these.



Take the **BearingBar.stl** (from **BearingColumns.stl**) 3D print and insert a 625-2RS into it as shown here.

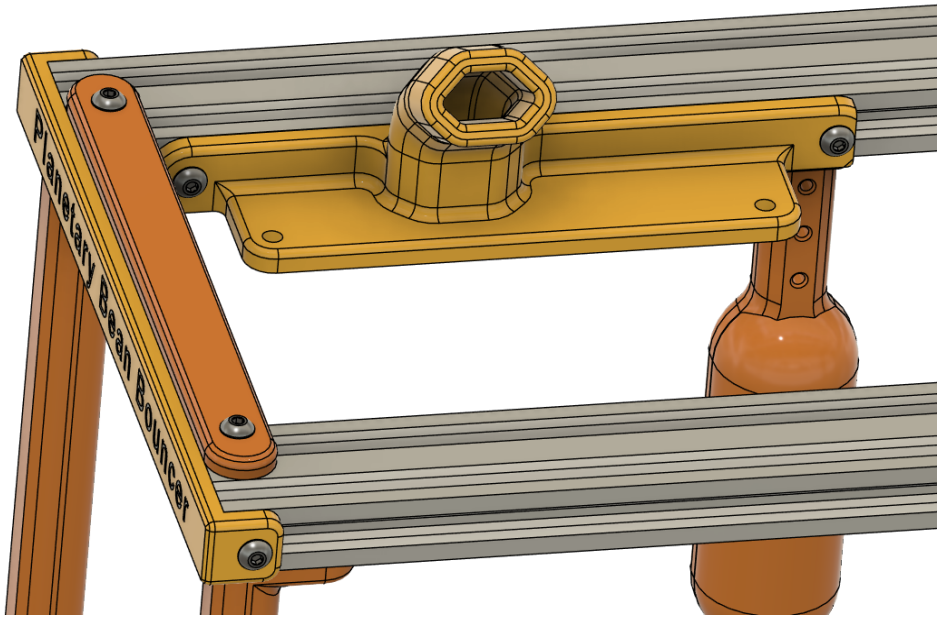


I created a plate of the STL files for this assembly. I count 25-M5x12mm screws, 23-M5 T-Nuts and 2-M5 Locknuts. The light blue part has a place for another screw, but not really needed since the back profile matches V-Slot and holds well enough with that.



Add the **ShortV_Connect_V3.stl** before putting the end piece on. This is used for connecting the Electronics case to the extrusion. If you are putting together the Arduino UNO case I made, you may want to attach the case and LCD case to this before attaching to the 2040 extrusion.

The next part to install is the **V2020x140_Mod.stl** 3D print. You could also use a 2020x140mm length of extrusion, but this uses only 1 Tee Nut on each side and you do not have any grooves for chaff to get in.



Except for putting the end caps on this completes the top frame construction. I would not put them on until the end in case you need to insert some Tee Nuts somewhere you missed. Only the 2 Tee Nuts in the short 2040x140mm extrusion need to be in there at this point. The others are easy to put in when you need them until the endcaps go on.

Optional 5:1 Planetary Gear

Nema 17 Motor Mount Assembly

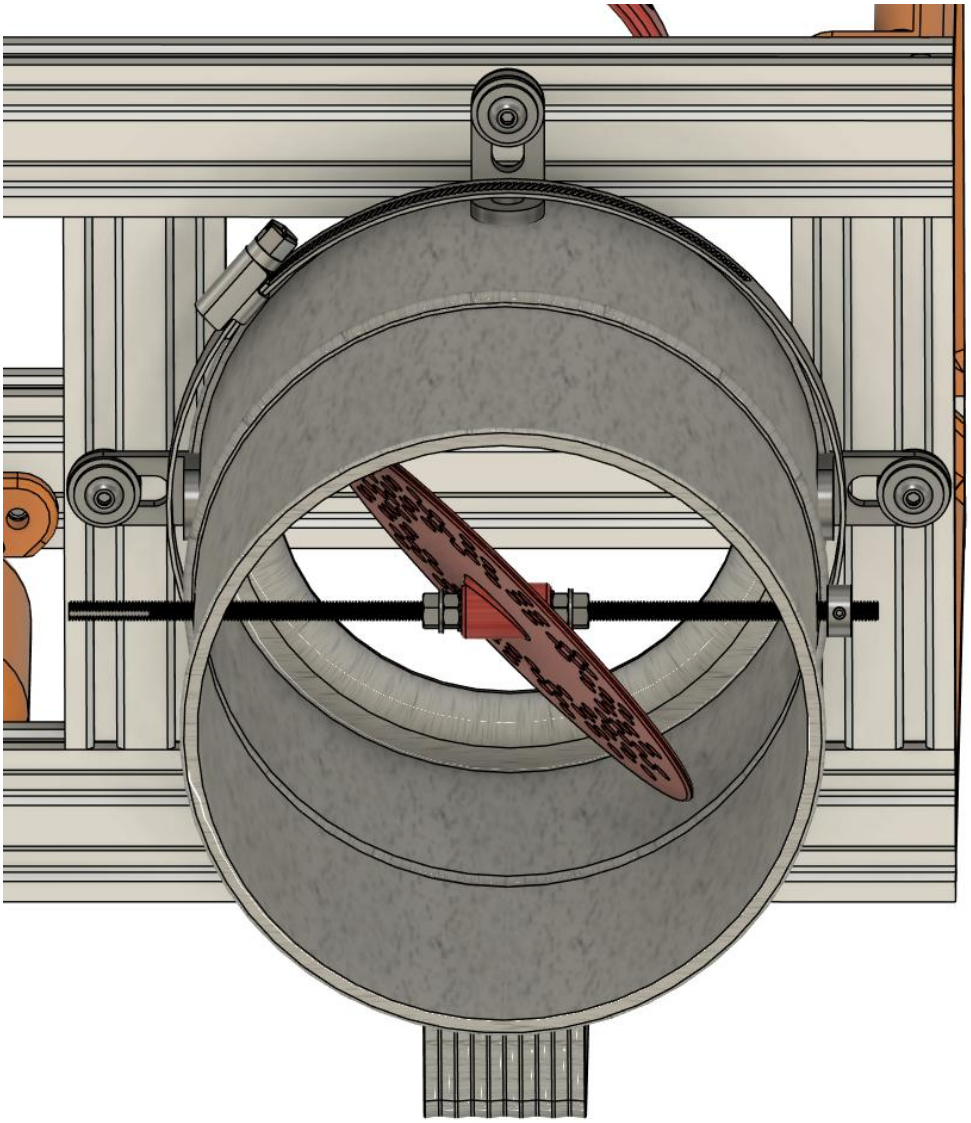
Wobble Disc Assembly with shaft

If you cut your own disc out of stainless steel sheet, your hardware will be different. Whatever method you use, I would recommend 3d printing **WobbleDisck_Mount_JamNuts.stl** out of PLA first to check the diameter size. Print it on a 45 degree angle and add manual tree support between bottom and mid-section. My disc ended up being about 5.14" or 130.175mm which is the size of that .stl file.

Hardware:

1 – 316L 3D Printed Stainless Steel Wobble disc
1 – M5x200mm fully threaded 18-8 stainless steel shaft (18-8 = 304 SS)
2 – 18-8 Stainless Steel External Tooth Lock Washers
4 – Regular M5 hex nuts (4mm thick to better get wrench on)
4 – M5 Serrated Flange Lock Nuts.
2 – M5x10x18mm 18-8 Stainless Steel Spacers

Only the 2 external tooth lock washers, 2 hex nuts, and 2 serrated flange lock nuts are really necessary. I added the spacers with hex nut, flange nut to cover the threads more. Using more spacers will make it more difficult to put the shaft in. I also recommend drilling out the shaft openings with a 3/8" cobalt drill bit so the shaft does not ride on the sifter at all. That larger hole also allows you to thread all the hardware on outside the sifter. I left about 15mm of the shaft on the outside bearing side and 31mm on the drive side. The coupling on the drive shaft side has 2 set screws 180 degrees apart, so I also filed down the shaft on the drive side on 2 sides so coupling gets better grip.

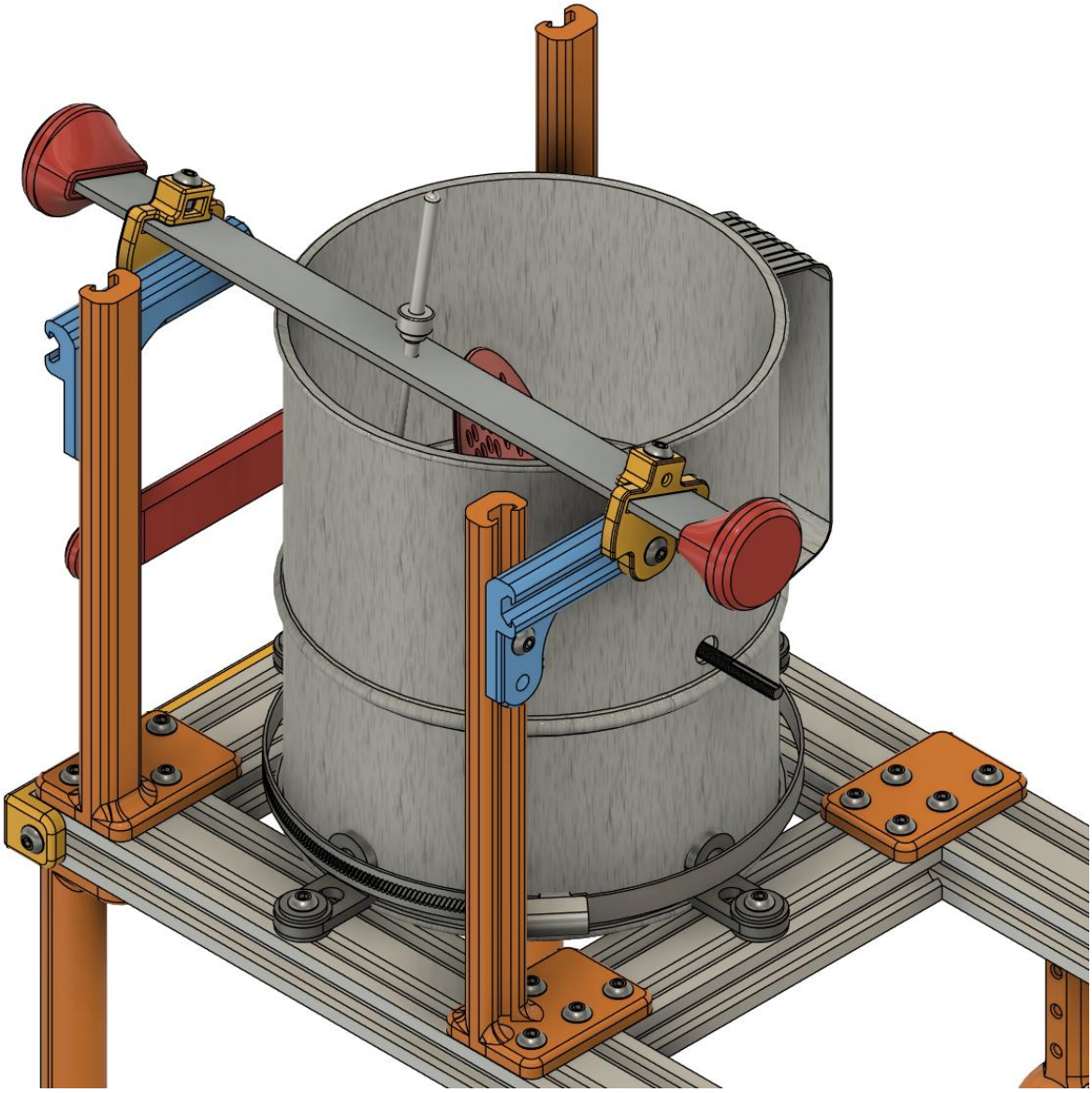


Mount Flour Sifter Assembly to Frame

Hardware:

1 – Assembly from previous procedure
1 - 5"-7" SS Band Clamp
4 – 18x18x30 L Brackets as shown in BOM
4 – M5x12mm screws
4 – Tee Nuts
12 - M5 fender washers as shown in BOM

Mount as shown in graphic below.

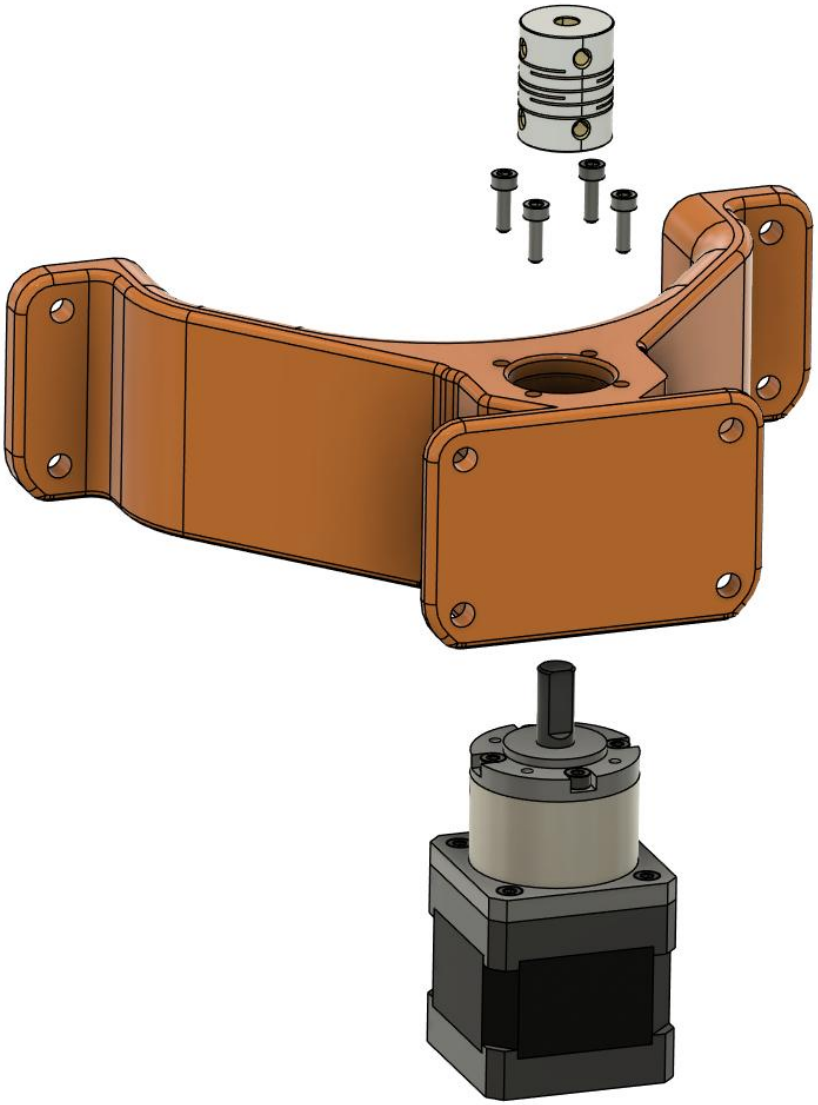


Mount Planetary gear assembly and Motor Mount Assembly

Mount Motor to Mount

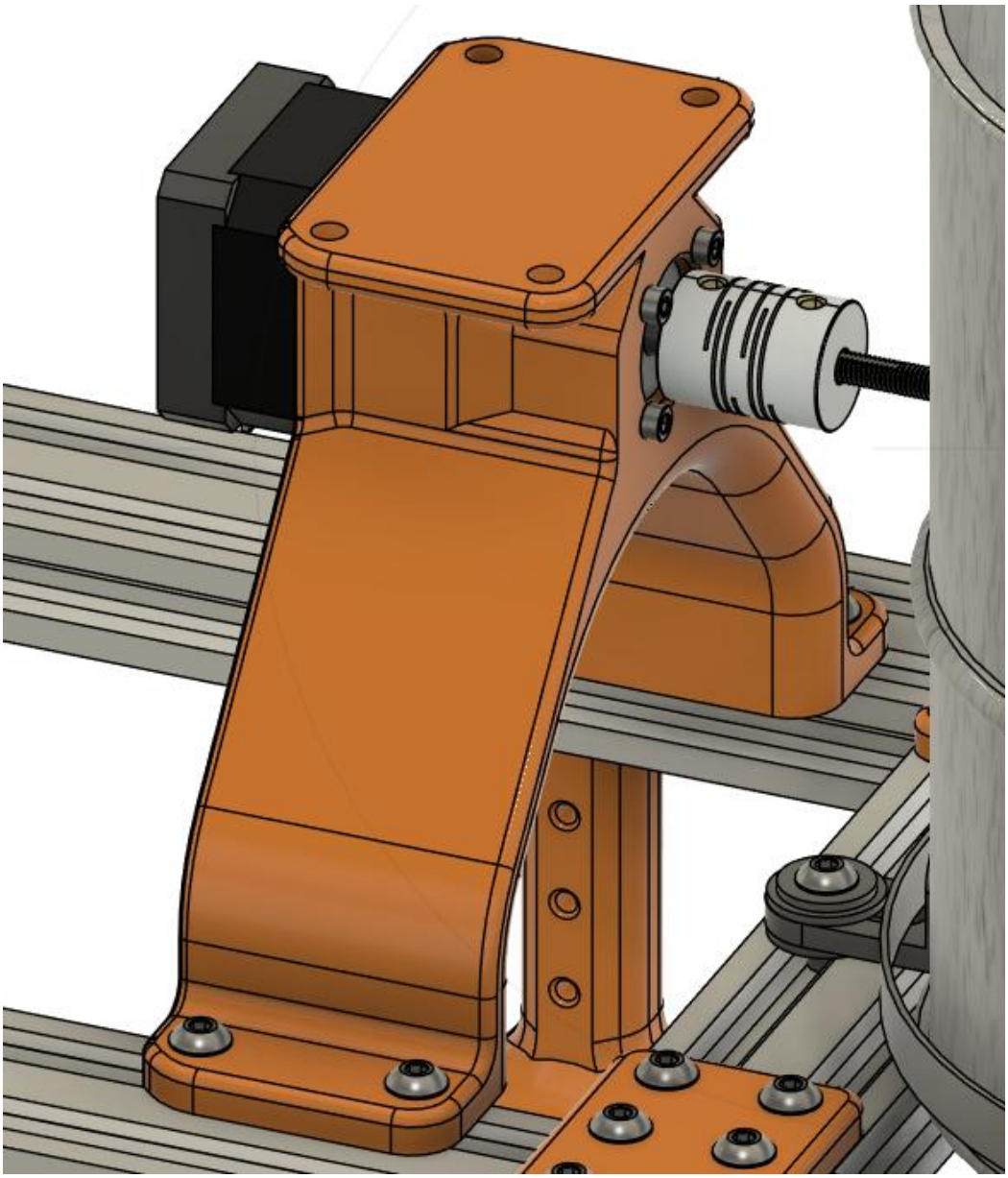
Hardware:

1 – 3D printed Nema17_PlanetMount_V2
1 – Nema 17 motor
1 - 8mm to 6mm Coupling Flexible Coupler Motor Connector
4 – M3x10mm screws
4 – M5x12mm screws
4 – M5 T-Nuts



Insert the Nema 17 Stepper Motor Geared Planetary Gearbox into the 3D printed Mount and screw in the 4 – M3x10mm screws to secure. Next add the coupler to the shaft end of the motor.

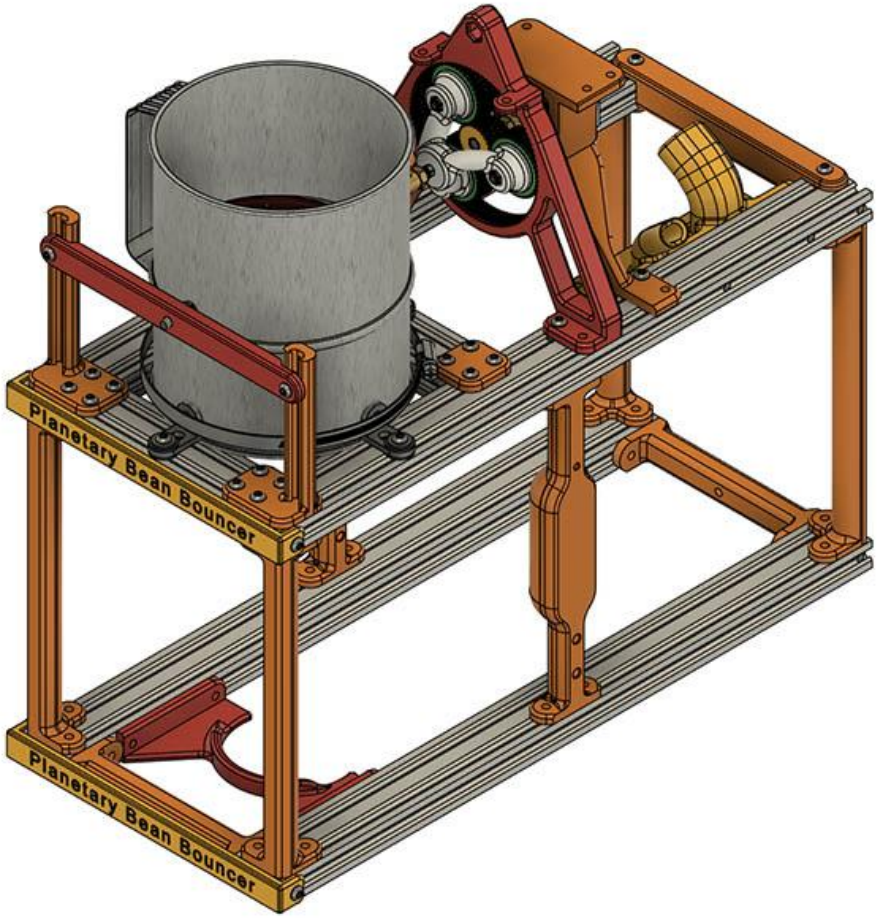
Now add the planetary gear assembly and motor mount assembly to the frame using 4 – M5x12mm screws and 4 – M5 T-Nuts.



End Cap Assembly

4 – 3D printed End caps
8 – M5x12mm screws
8 – M5 Tee Nuts.

Add the End Caps as shown. They will be the same on both ends



This completes the frame and mechanical build.